

SpeechEditBench: A Bilingual Multi-Attribute Benchmark for Instruction-Guided Speech Editing

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Can a model make **exactly the requested edits** while preserving **everything else**?

Why SpeechEditBench?

Existing evaluations are **fragmented by task** and often score only whether the requested change happened.

Editing is a **dual-constraint** problem:
change the target · **preserve the rest**

1 Unified coverage

7 atomic tasks plus 2- and 3-way compositional editing.

2 Bilingual diagnosis

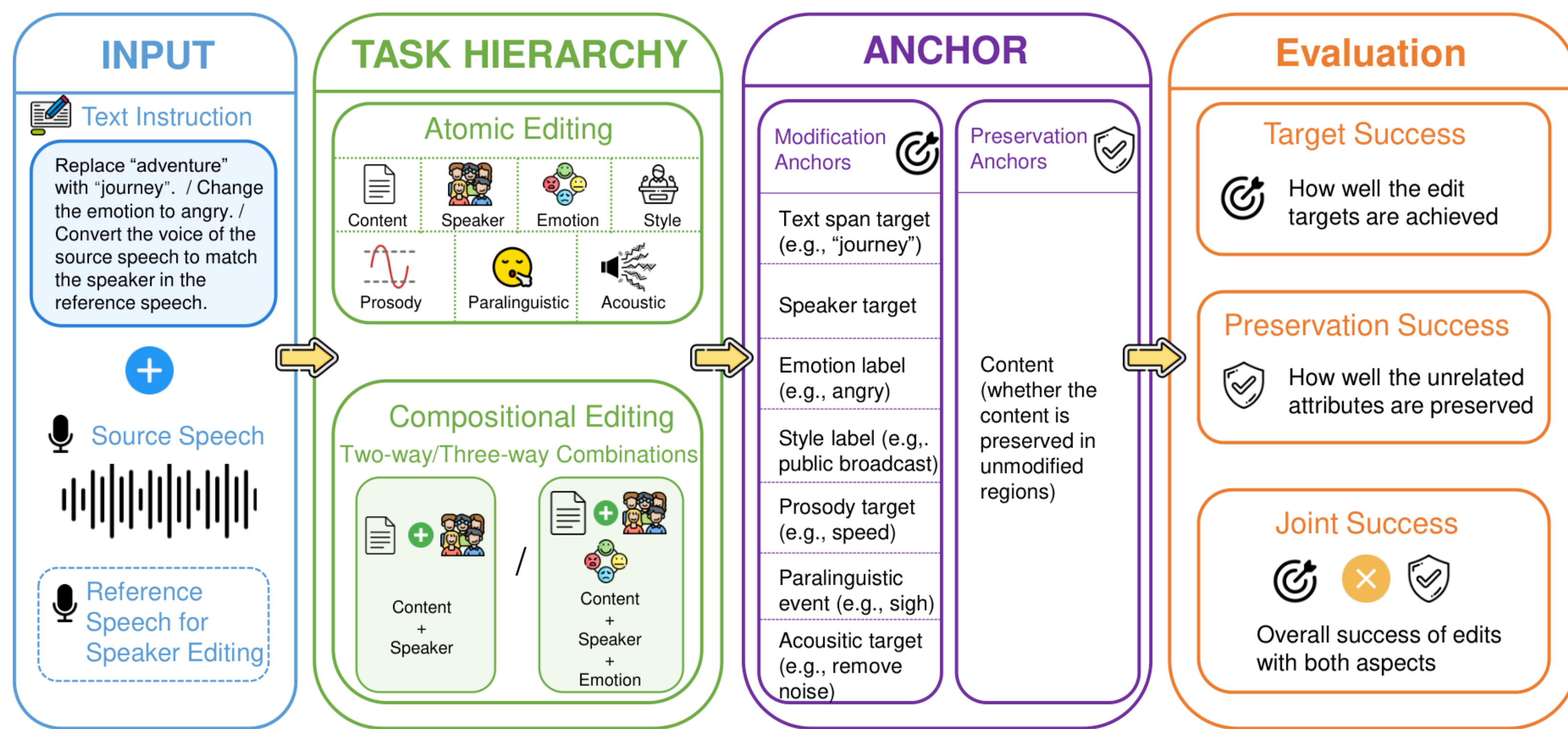
Balanced English and Mandarin data reveal model-dependent language bias.

3 Anchor-based metrics

Separate target success, preservation, and their conjunction.

Benchmark overview

Natural-language instruction + source speech → task-specific anchors → target, preservation, and joint success



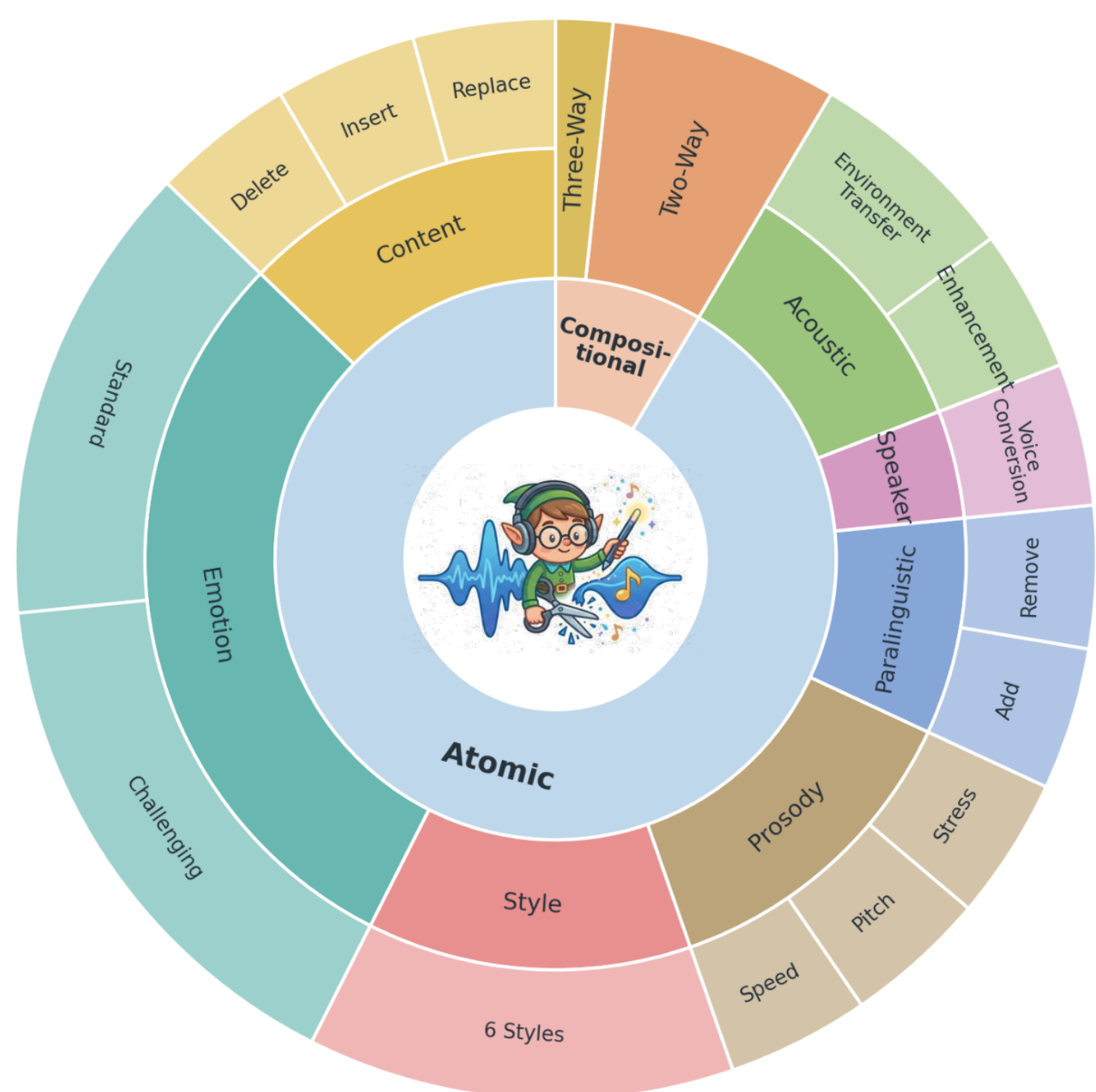
Dataset at a glance

4,700
source-instruction pairs

4,300 + 400
atomic + compositional

2,400 / 2,300
English / Mandarin

320 + 80
2-way / 3-way edits



Task distribution

Content 600	Speaker 200	Emotion 1,400	Style 600
Prosody 600	Paralinguistic 400	Acoustic 500	Compositional 400

Task → primary evaluator

Content	target span → ASR alignment
Speaker	reference voice → cosine ≥ .50
Emotion	label → multimodal judge
Style	label → target score ≥ 3
Prosody	range → duration / F0 / prominence
Paralinguistic	event → add / remove score
Acoustic	condition → DNSMOS / RT60 / PANNs
Compositional	components → all + joint success

Each sample uses task-specific anchors—not a single rigid reference waveform.

Anchor-based evaluation

t_i
Target
success

p_i
Preservation
success

j_i
Joint
success

$$j_i = t_i \times p_i$$

Modification anchors: target span, speaker reference, emotion/style label, prosody range, event, acoustic condition.
Preservation gate: expected transcript WER/CER ≤ 10%.

Target vs. joint capability

Joint profiles contract when content preservation is enforced.



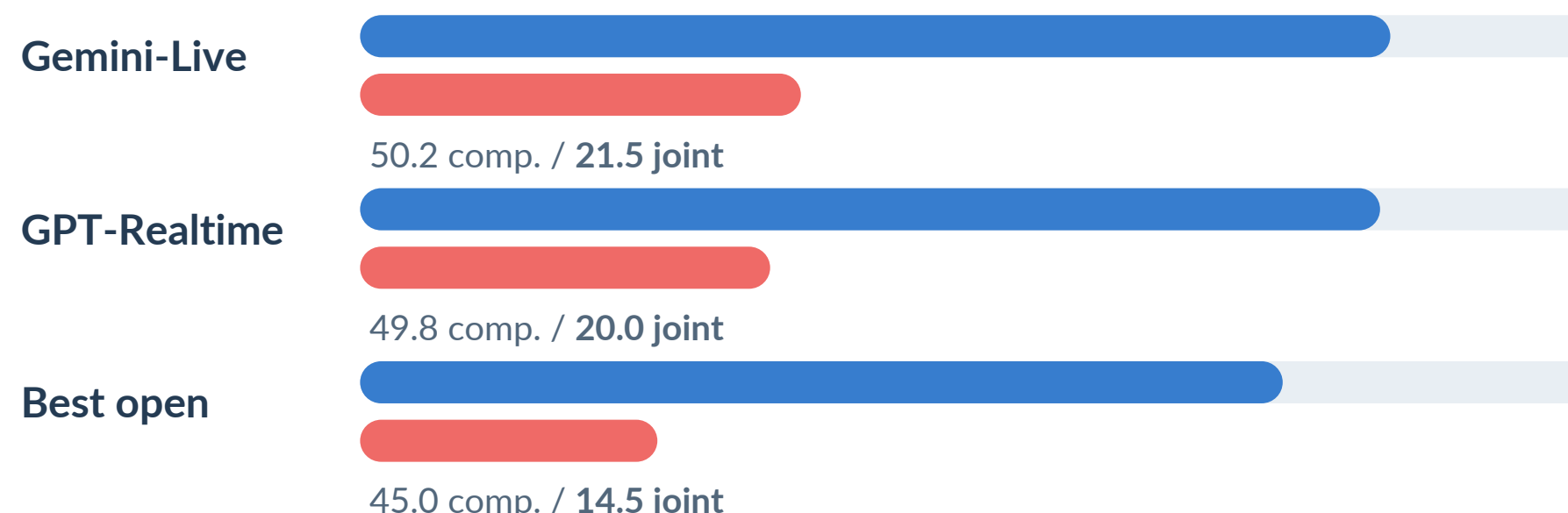
Axes are normalized by the best Speech LLM score per task.

Compositional challenge

2-component, speaker-free subset

■ Component success

■ Joint success



3-component: nearly every model reaches **0% joint success**.

Three findings

- 01 No universal winner — Strengths are task-specific.
- 02 Closed models lead — But Qwen3-Omni wins acoustic editing.
- 03 Preservation bottleneck — Target success can hide content corruption.

Best Speech LLM joint success

Best model varies by task → capability is fragmented.

Content	96.7 GPT-Realtime
Emotion	27.8 Gemini-Live
Style	68.7 GPT-Realtime
Prosody	65.2 Gemini-Live
Paraling.	47.0 GPT-Realtime
Acoustic	37.8 Qwen3-Omni
Compos.	11.0 Gemini-Live

closed-source open-source

Human validation

68.8%
agreement

240
outputs

720
judgments

Joint success · 95% CI 62.6–74.3
Pearson .685 · Spearman .699 (both p<.001)

Takeaway

Edit success alone is not enough.
Future speech editors must satisfy the instruction *and* preserve untargeted content—especially under multi-attribute commands.



SCAN FOR DATA + CODE

github.com/daxintan-cuhk/SpeechEditBench

arXiv:2606.01804 · Apache-2.0 code and benchmark assets

Questions? → tan.daxin1@huawei.com

SpeechEditBench v1.1 · 4,700 samples · English + Mandarin